جامعة حلوان – كلية الهندسة
قسم هندسة الحاسبات والنظم

Helwan University – Faculty of Engineering
Department of Computer and Systems Engineering

UX Phase II and III

Smart LMS with AI Proctoring

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1. Problem Statement and Ideation:

A. Problem Statement

Educational institutions currently face a massive gap in digital transformation. Traditional systems are often fragmented, where student data, course materials, and examination security are handled by separate, disconnected platforms. This leads to:

- **Educational Inefficiency**: Managing levels, specific courses, assignments, and quizzes manually across different years is prone to human error.
- **Security & Integrity Gaps**: The absence of a unified, secure authentication and proctoring system makes it difficult to verify student identities and ensure exam honesty.
- **Poor User Engagement**: Students lack a single hub to track their sheets, assignments, and academic progression in one place.

B. Ideated Solution: The Lumina Ecosystem

Lumina LMS was ideated as an all-in-one educational ecosystem that automates the entire academic lifecycle. The system is designed to be a complete, ready-to-use solution that handles:

- **Automated Level Management**: The system intelligently categorizes students into academic levels, where each level automatically unlocks its specific set of courses, assignments, and sheets.
- **Comprehensive Assessment Suite**: A fully integrated engine for managing quizzes, assignments, and formal examinations tailored to each academic stage.
- **AI-Powered Proctoring & Security**: A built-in, native proctoring system that monitors exams in real-time to prevent cheating, combined with JWT-protected security for all user data and profile management.
- **Unified Student Hub**: A modern, responsive dashboard where students can manage their identity, track their 'Level Up' progress, and access all their educational requirements through a seamless interface.

C. Strategic Value

The goal is to provide a 'Zero-Friction' environment. For the institution, it's a tool for absolute control and integrity. For the student, it's a clear, automated path toward graduation where every course, sheet, and exam is organized within a high-security framework.

2. Wireframe and Interactions:

System Layout & Architecture

The wireframe for **Lumina LMS** is structured to provide a high-efficiency educational environment. It consists of four primary interconnected screens designed with a **'Flat UI'** philosophy to minimize user navigation effort:

1. **Identity & Access Portal (Login/Register)**: A secure, minimalist interface focused on user authentication and JWT token generation.
2. **Level-Based Global Dashboard**: The central hub that dynamically serves content (Courses, Assignments, Sheets) based on the student's verified academic year.
3. **The Interactive Identity Drawer**: A sleek, side-sliding panel that handles all personal profile modifications, security key updates, and academic 'Level-Up' requests.
4. **AI Proctoring Exam Environment**: A specialized high-security layout that integrates a live camera feed overlay to monitor integrity during assessments.

3. Design:

A. Project Colors (utils/colors.js)

To maintain a consistent brand identity, we have established a professional color palette gathered in the utils folder:

- **Primary Color**: #41B3E3 (Lumina Light Blue) – Used for primary action buttons, active navigation states, and the profile border.
- **Secondary Color**: #002D5B (Lumina Dark Navy) – Used for headers, typography, and professional branding elements.
- **Neutral/Background**: #FFFFFF & #F8F9FA – Utilized for glass-card backgrounds and minimalist dashboard layouts to ensure high readability.
- **Danger/Alert Color**: #FF4B2B – Reserved for security settings and critical actions."

B. Project Fonts (assets/fonts)

Standardized typography is stored within the assets folder to ensure consistent rendering across all platforms:

- **Title & Headers**: **Montserrat (Bold)** – A modern, geometric font chosen for dashboard titles and section headings.
- **Regular & Body Text**: **Poppins (Regular)** – A highly legible sans-serif font used for form fields, assignment descriptions, and general text.

C. Custom Buttons (components/ folder)

The following custom buttons were designed to provide a high-end, interactive user experience throughout the **Lumina LMS** platform:

- **btn-levelup (Academic Progression Button)**:
    - **Purpose**: Used to trigger the logic for advancing the student to the next academic year.

    - **Styling**: Features a linear gradient from Lumina Light Blue (#41B3E3) to Dark Navy (#002D5B) with an upward-pointing icon (bi-arrow-up-circle).
    - **Interaction**: Includes a hover elevation effect and a distinct box-shadow for a 'clickable' feel.
- **btn-login-grad (Primary Action Button)**:
    - **Purpose**: The main button for submitting forms, such as 'Save Changes' in the profile info section.
    - **Styling**: Utilizes the brand's primary gradient and bold white typography to ensure high visibility.
- **btn-danger-grad (Security Action Button)**:
    - **Purpose**: Used for high-security actions like 'Update Password'.
    - **Styling**: A warm-to-hot gradient (#ff4b2b to #ff416c) that signifies a critical security update.
- **btn-outline-info & btn-info-active (Navigation Toggles)**:
    - **Purpose**: Used for the accordion headers to toggle the 'Edit Profile' section.
    - **Styling**: Starts as an outlined button and transitions to a solid blue background when active or hovered, ensuring the student knows which section is open.
- **btn-outline-danger & btn-danger-active (Security Toggles)**:
    - **Purpose**: Used to toggle the 'Change Security Key' section.
    - **Styling**: Follows the same logic as the info toggle but uses the danger red color scheme to distinguish security tasks.
- **edit-photo-btn (Media Management Button)**:
    - **Purpose**: A floating action button (FAB) located on the profile picture for uploading new images.
    - **Styling**: A circular blue button containing a camera icon (bi-camera-fill), positioned absolutely over the user's avatar.
- **logout-btn (Session Management)**:
    - **Purpose**: Located in the top navigation bar to securely terminate the JWT session.

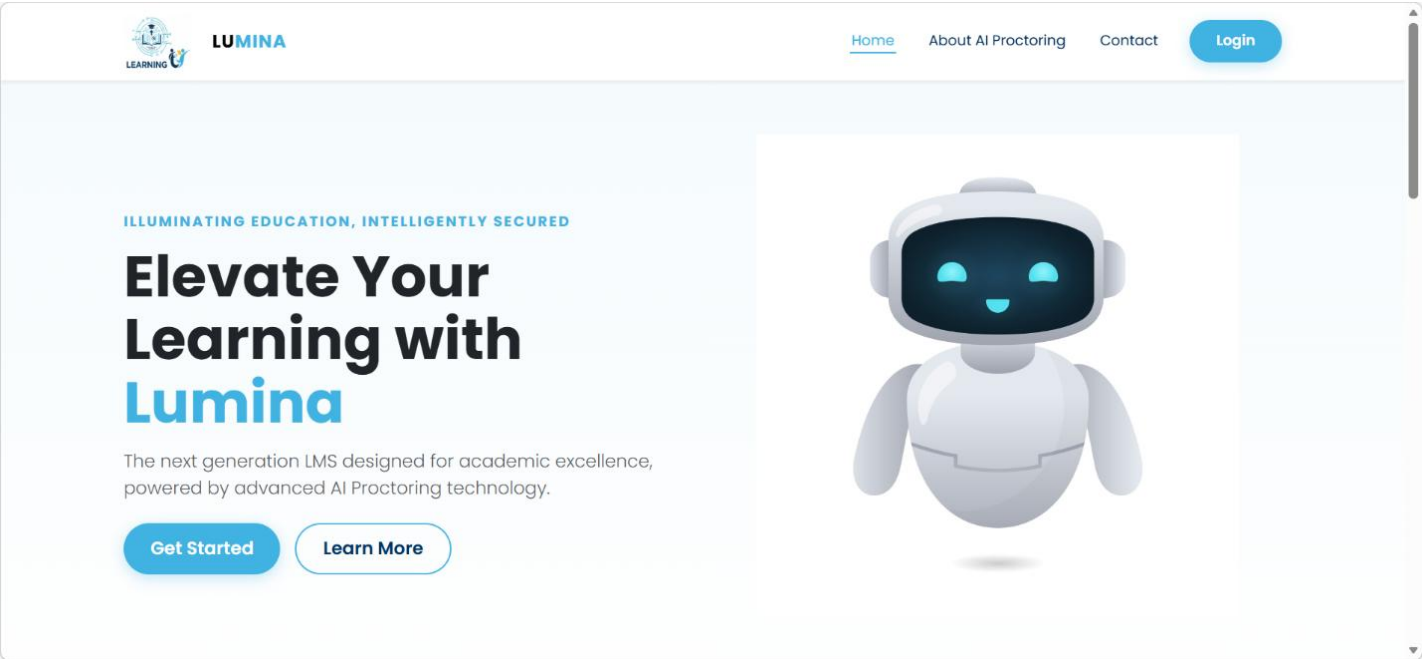
- **Styling**: A minimalist, clean button with a subtle border, designed to be accessible but not distracting from main tasks.

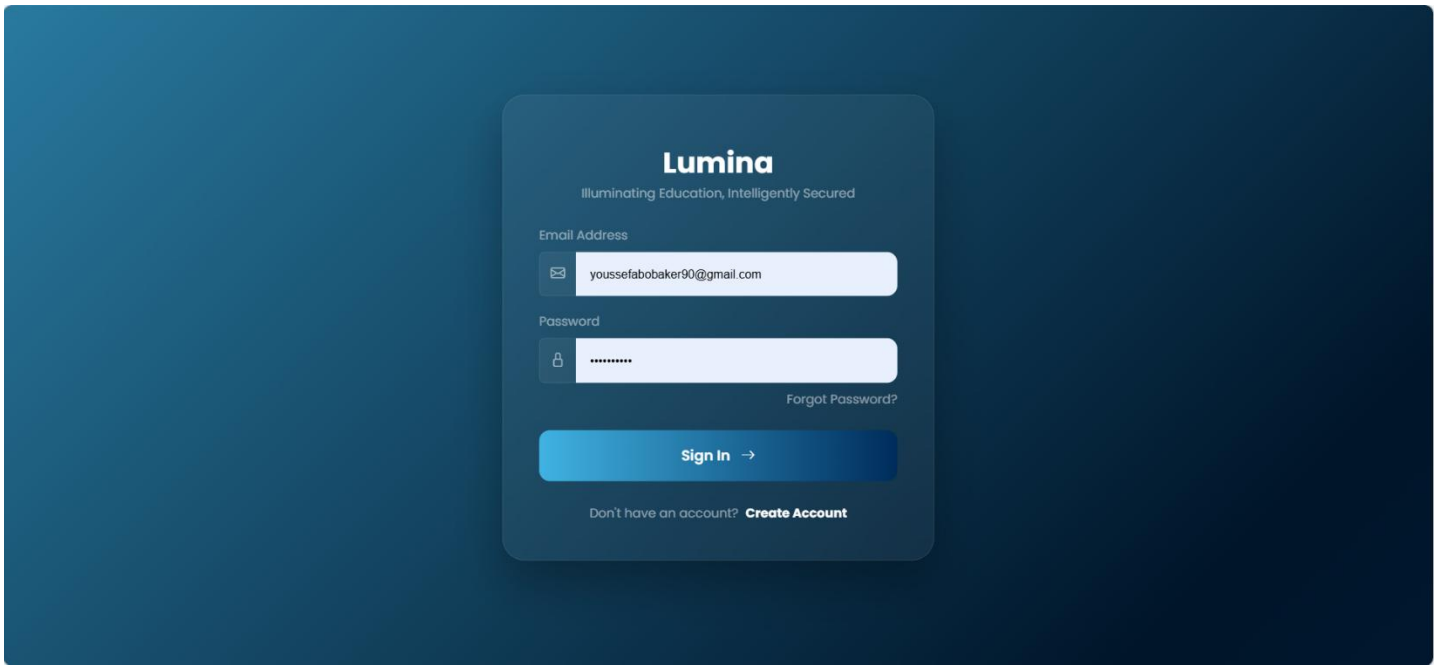
D. Initial Screens (screens/ folder)

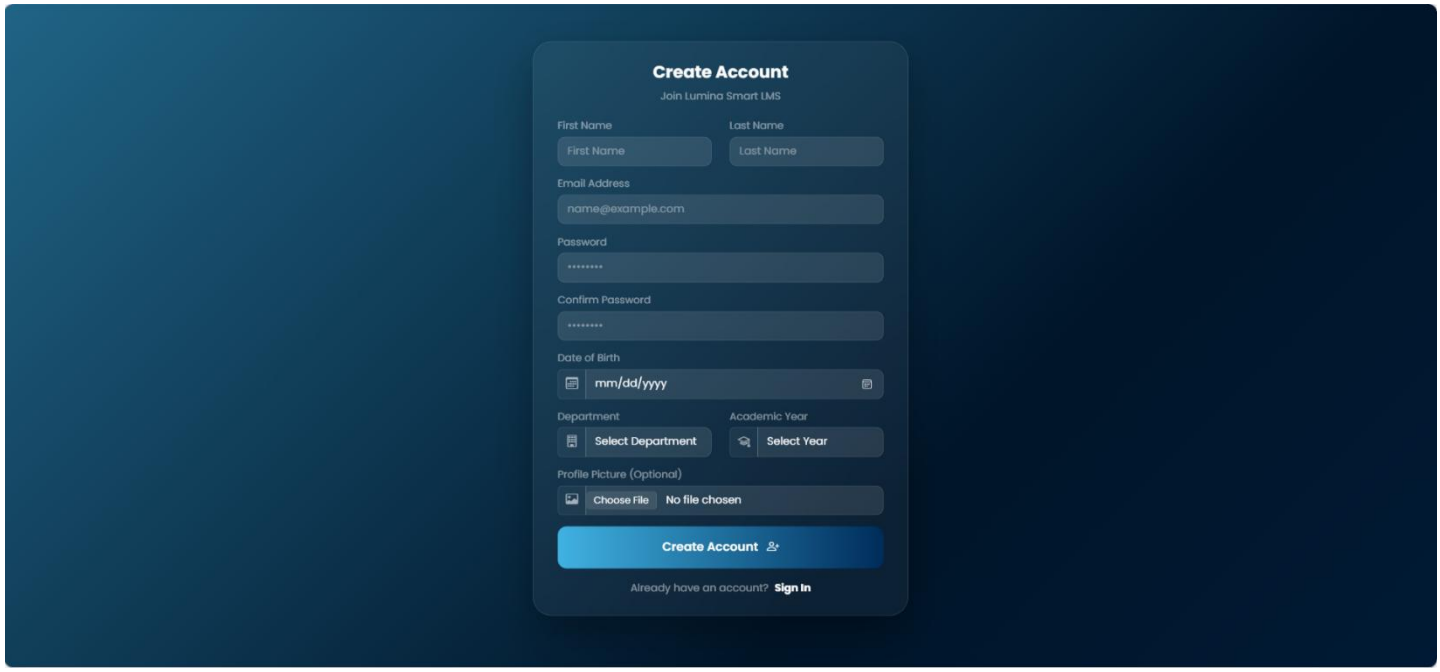
The project's primary views are organized within a screens folder in the main root for clear architectural separation:

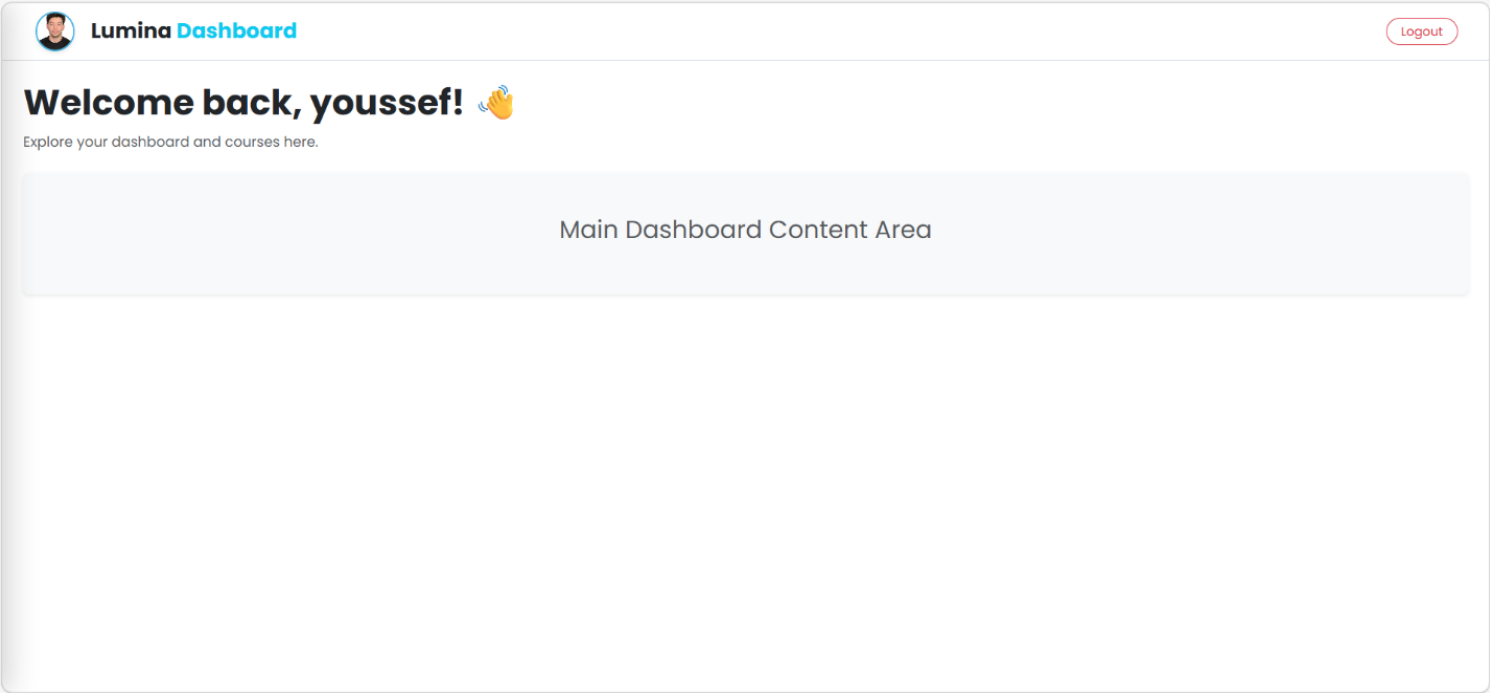
- **AuthScreen**: Consolidates the Login and Registration layouts with secure JWT integration.
- **DashboardScreen**: The primary hub where students view courses, sheets, and assignments filtered by their academic level.
- **ExamProctoringScreen**: A high-security screen featuring the AI monitoring HUD and live camera feed during assessments.

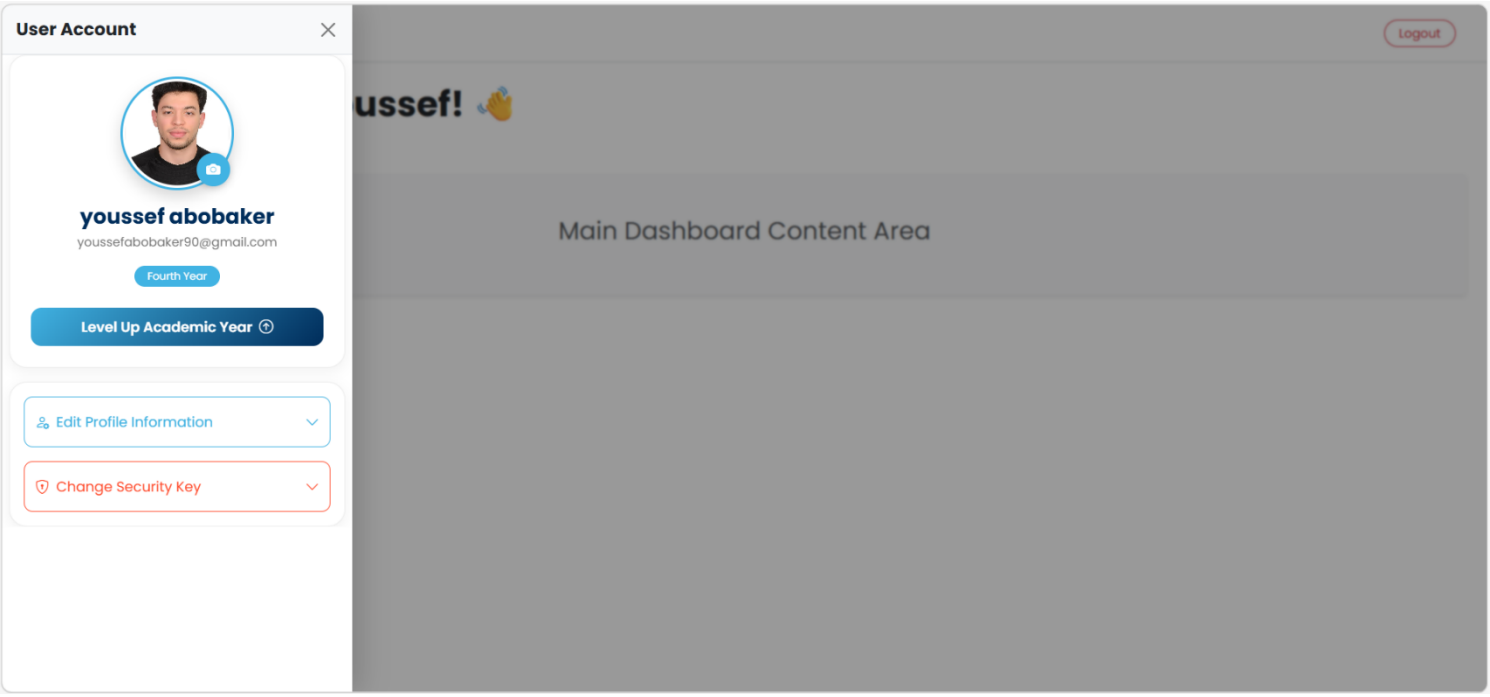
4. Project Implementation Screenshots:




Lumina
Illuminating Education, Intelligently Secured
Email Address
youssefabobaker90@gmail.com
Password
Forgot Password?
Sign In
Don't have an account? Create Account


Create Account
Join Lumina Smart LMS
First Name
Last Name
First Name
Last Name
Email Address
name@example.com
Password
Confirm Password
Date of Birth
mm/dd/yyyy
Department
Academic Year
Select Department
Select Year
Profile Picture (Optional)
Choose File
No file chosen
Create Account
Already have an account? Sign In


Lumina Dashboard
Logout
Welcome back, youssef! 👋
Explore your dashboard and courses here.
Main Dashboard Content Area


User Account
youssef abobaker
youssefabobaker90@gmail.com
Fourth Year
Level Up Academic Year
Edit Profile Information
Change Security Key
Logout
ussef! 👋
Main Dashboard Content Area